

```
In [1]:
import pandas as pd
```

```
In [2]:
import numpy as np
```

```
In [3]:
import seaborn as sns
```

```
In [4]:
import matplotlib.pyplot as plt
```

```
In [5]:
df=pd.read_csv(r'C:\Users\M A H N A Z\Downloads\customer_shopping_behavior.csv')
```

```
In [6]:
df.head()
```

Out[6]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	4.5
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	4.0
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	4.5
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	4.0
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	4.5

```
In [7]:
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   str
3   Item Purchased                      3900 non-null   str
4   Category                            3900 non-null   str
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                            3900 non-null   str
7   Size                                3900 non-null   str
8   Color                               3900 non-null   str
9   Season                              3900 non-null   str
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   str
12  Shipping Type                      3900 non-null   str
13  Discount Applied                   3900 non-null   str
14  Promo Code Used                    3900 non-null   str
```

15 Previous Purchases 3900 non-null int64
16 Payment Method 3900 non-null str
17 Frequency of Purchases 3900 non-null str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB

In [8]:

```
df.describe(include='all')
```

Out[8]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN

In [9]:

```
df.isnull().sum()
```

Out[9]:

Customer ID 0
Age 0
Gender 0
Item Purchased 0
Category 0
Purchase Amount (USD) 0
Location 0
Size 0
Color 0
Season 0
Review Rating 37
Subscription Status 0
Shipping Type 0
Discount Applied 0
Promo Code Used 0
Previous Purchases 0
Payment Method 0
Frequency of Purchases 0
dtype: int64

In [10]:

```
df['Review Rating']=df.groupby('Category')['Review Rating'].transform(lambda x:x.fillna(
```

In [11]:

```
df.isnull().sum()
```

Out[11]:

```
Customer ID          0
Age                  0
Gender               0
Item Purchased       0
Category             0
Purchase Amount (USD) 0
Location            0
Size                0
Color               0
Season              0
Review Rating        0
Subscription Status  0
Shipping Type        0
Discount Applied     0
Promo Code Used      0
Previous Purchases   0
Payment Method       0
Frequency of Purchases 0
dtype: int64
```

In [16]:

```
df.columns=df.columns.str.lower()
df.columns=df.columns.str.replace(' ','_')
df=df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

In [17]:

```
df.columns
```

Out[17]:

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

create a new column

In [18]:

```
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] =pd.qcut(df['age'], q=4, labels = labels)
```

In [19]:

```
df[['age', 'age_group']].head(10)
```

Out[19]:

```
   age  age_group
0   55  Middle-aged
```

	age	age_group
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

create purchase frequency days

In [20]:

```
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

In [23]:

```
df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

Out[23]:

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

In [24]:

```
df[['discount_applied', 'promo_code_used']].head(10)
```

Out[24]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

In [26]:

```
(df['discount_applied'] == df['promo_code_used']).all()
```

Out[26]:

np.True_

In [27]:

```
df=df.drop('promo_code_used', axis=1)
```

In [28]:

```
df.columns
```

Out[28]:

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
      'purchase_amount', 'location', 'size', 'color', 'season',  
      'review_rating', 'subscription_status', 'shipping_type',  
      'discount_applied', 'previous_purchases', 'payment_method',  
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],  
      dtype='str')
```

In [29]:

```
pip install pycpg2-binary sqlalchemy
```

Collecting pycpg2-binary

Downloading pycpg2_binary-2.9.12-cp314-cp314-win_amd64.whl.metadata (5.1 kB)

Collecting sqlalchemy

Downloading sqlalchemy-2.0.50-cp314-cp314-win_amd64.whl.metadata (9.8 kB)

Collecting greenlet>=1 (from sqlalchemy)

Downloading greenlet-3.5.1-cp314-cp314-win_amd64.whl.metadata (3.9 kB)

Requirement already satisfied: typing-extensions>=4.6.0 in C:\Users\M A H N A Z\AppData\Local\Programs\Python\Python314\Lib\site-packages (from sqlalchemy) (4.15.0)

Downloading pycpg2_binary-2.9.12-cp314-cp314-win_amd64.whl (2.8 MB)

```
----- 0.0/2.8 MB ? eta -:-:--  
----- 0.8/2.8 MB 9.1 MB/s eta 0:00:01  
----- 1.6/2.8 MB 4.8 MB/s eta 0:00:01  
----- 2.1/2.8 MB 3.7 MB/s eta 0:00:01  
----- 2.8/2.8 MB 4.3 MB/s 0:00:00
```

```

Downloading sqlalchemy-2.0.50-cp314-cp314-win_amd64.whl (2.1 MB)
----- 0.0/2.1 MB ? eta -:-:-
----- 0.8/2.1 MB 4.2 MB/s eta 0:00:01
----- 1.6/2.1 MB 4.0 MB/s eta 0:00:01
----- 2.1/2.1 MB 3.9 MB/s 0:00:00
Downloading greenlet-3.5.1-cp314-cp314-win_amd64.whl (239 kB)
Installing collected packages: psycpg2-binary, greenlet, sqlalchemy

```

Successfully installed greenlet-3.5.1 psycopg2-binary-2.9.12 sqlalchemy-2.0.50

In [30]:

```
Cell In[30], line 1
python -m pip install --upgrade pip
      ^
SyntaxError: invalid syntax
```

```
%pip install --upgrade pip
```

```

Downloading pip-26.1.2-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.1.2-py3-none-any.whl (1.8 MB)
----- 0.0/1.8 MB ? eta -:-:-
----- 0.3/1.8 MB ? eta -:-:-
----- 1.0/1.8 MB 6.3 MB/s eta 0:00:01

```

----- 1.8/1.8 MB 4.7 MB/s 0:00:00
Installing collected packages: pip
 Attempting uninstall: pip
 Found existing installation: pip 26.0.1
 Uninstalling pip-26.0.1:
 Successfully uninstalled pip-26.0.1
Successfully installed pip-26.1.2
Note: you may need to restart the kernel to use updated packages.

In [32]:

```
pip install psycopg2-binary sqlalchemy
```

Requirement already satisfied: psycopg2-binary in C:\Users\M A H N A Z\AppData\Local\Programs\Python\Python314\Lib\site-packages (2.9.12)
Requirement already satisfied: sqlalchemy in C:\Users\M A H N A Z\AppData\Local\Programs\Python\Python314\Lib\site-packages (2.0.50)
Requirement already satisfied: greenlet>=1 in C:\Users\M A H N A Z\AppData\Local\Programs\Python\Python314\Lib\site-packages (from sqlalchemy) (3.5.1)
Requirement already satisfied: typing-extensions>=4.6.0 in C:\Users\M A H N A Z\AppData\Local\Programs\Python\Python314\Lib\site-packages (from sqlalchemy) (4.15.0)
Note: you may need to restart the kernel to use updated packages.

In [35]:

```
# Step 1: Connect to PostgreSQL  
# Replace placeholders with your actual details  
from sqlalchemy import create_engine  
username = "postgres"  
password = "Mahi2001#" # the password you set during installation  
host = "localhost"  
port = "5432"  
database = "customer_behavior"  
  
engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")  
  
# Step 2: Load DataFrame into PostgreSQL  
table_name = "customer" # choose any table name  
df.to_sql(table_name, engine, if_exists="replace", index=False)  
  
print(f"Data successfully loaded into table '{table_name}' in database ' {database}'.")
```

Data successfully loaded into table 'customer' in database ' customer_behavior'.

In []: